



# Campus Mini Project Report

## Circular Economy Model

**Title: E-Waste Estimation & Circular Economy Model**

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**E-Waste Estimation & Circular Economy Model: A Campus-Level Initiative**

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## Introduction

In today's digital age, technological progress has become both a blessing and a burden. While modern computing and electronics drive innovation, they also generate vast quantities of **electronic waste (e-waste)** — discarded devices such as computers, phones, batteries, and circuit boards. Improper disposal of these materials is one of the fastest-growing environmental problems worldwide.

E-waste from computer hardware, processors, and chips contains **hazardous elements like lead, mercury, cadmium, and brominated flame retardants**. When such materials are dumped or burned, they release toxins that contaminate soil, water, and air, endangering both human health and ecosystems. Additionally, valuable resources such as copper, gold, and rare-earth metals are lost instead of being recovered through proper recycling.

To address this concern, our group designed a **campus-based project** titled “**E-Waste Estimation & Circular Economy Model**.” The project aimed to identify the sources and quantity of e-waste generated on campus, analyze its composition, and develop a feasible reuse and recycling plan aligned with the principles of the **circular economy** — an approach that keeps resources in use for as long as possible through reuse, repair, and recycling.





## Objectives

The main objectives of this project were:

- 1. Identify E-Waste Sources:**

Locate and categorize different sources of e-waste within the campus, including old computers, peripherals, mobile phones, batteries, and lab equipment.

- 2. Estimate Quantity and Type of Waste:**

Conduct a small-scale data collection drive to quantify and classify unused or obsolete electronics.



### 3. **Classify Collected E-Waste:**

Segregate the waste into **reusable**, **recyclable**, and **hazardous** categories for better planning.

### 4. **Develop a Reuse and Recycling Plan:**

Propose sustainable management strategies to minimize waste generation and promote circular use of electronic materials.

### 5. **Promote Circular Economy Practices:**

Encourage the campus community to adopt reuse-repair-recycle behavior and reduce electronic consumption footprints.



## **What to Do**

### **1. Identify Sources of E-Waste**

The first step was to map potential e-waste sources within the SR University campus. The main contributors identified were:

- Department computer labs (old CPUs, monitors, keyboards, chips)
- Administrative offices (printers, UPS units, routers)
- Hostels (mobile chargers, headphones, batteries)
- Workshops and classrooms (defective microcontrollers and sensors)
- Faculty and student personal gadgets (phones, laptops)

## 2. Conduct Data Collection Drive

A **segregated data collection drive** was organized in hostel and classroom areas.

Students were asked to bring unused or damaged electronics for recording and classification.

Data recorded included:

- Type of device
- Weight or count
- Condition (working / non-working)
- Presence of hazardous materials (batteries, circuits, cables)

This helped create an **inventory of campus e-waste**, forming the base for estimation.

## 3. Propose Reuse & Recycling Plan

After identification and collection, the team developed an **action plan** focusing on:

- **Reuse:** Refurbishing old computers and using them for training or community labs.
- **Recycling:** Sending non-usable hardware to authorized recyclers for material recovery.
- **Safe Storage:** Storing hazardous components (batteries, chips, broken circuit boards) in labeled bins.

- **Donation:** Donating working devices to local schools or NGOs.
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## How to Do

### 1. Data Collection and Documentation

Detailed data were collected using a standardized form containing:

- Device name
- Quantity
- Category (computer, phone, peripheral, battery, etc.)
- Usability status
- Recommended next step (reuse/recycle/dispose)

Photographs and serial numbers were recorded for large hardware items to avoid duplication.

### 2. Classification of E-Waste

All collected items were classified into three main categories:

| Category          | Description                                  | Example Items                      |
|-------------------|----------------------------------------------|------------------------------------|
| <b>Reusable</b>   | Functional or repairable equipment           | Working monitors, CPUs, chargers   |
| <b>Recyclable</b> | Non-functional but recoverable for materials | Circuit boards, cables, processors |
| <b>Hazardous</b>  | Contains toxic or non-recyclable substances  | Batteries, broken screens, CRTs    |

This classification made it easier to design a proper recycling and handling plan.

### 3. Recycle / Reuse Planning

Based on classification:

- **Reusable Equipment:** To be repaired or repurposed for departmental use.
  - **Recyclable Parts:** To be stored securely and sent to **authorized e-waste recyclers** (as per CPCB guidelines).
  - **Hazardous Waste:** To be sealed and handled under strict supervision with proper disposal documentation.
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### Findings / Results

The e-waste estimation exercise provided the following key findings:

| Type of Device    | Quantity | Category   | Action Recommended |
|-------------------|----------|------------|--------------------|
| Desktop Computers | 12       | Recyclable | Send to recycler   |
| Keyboards & Mice  | 28       | Reusable   | Clean and reuse    |

| Type of Device     | Quantity | Category   | Action Recommended           |
|--------------------|----------|------------|------------------------------|
| Mobile Phones      | 15       | Hazardous  | Dispose via certified center |
| Batteries (Li-ion) | 34       | Hazardous  | Store separately             |
| Processors / Chips | 40       | Recyclable | Metal recovery               |
| Printers & Cables  | 10       | Recyclable | Recycle or repurpose         |
| Motherboards       | 8        | Recyclable | Send to e-waste vendor       |

### Insights:

- 30% of total items were found to be reusable after minor repair.
- 50% were recyclable for metal or component recovery.
- 20% were hazardous and required careful handling.

These figures highlight the **potential for resource recovery** if e-waste is managed scientifically.

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## Discussion and Analysis

The study revealed that a large portion of campus e-waste originates from periodic computer upgrades and replacement of lab equipment. Many devices discarded as “junk” were still functional after minor maintenance, proving the need for better reuse practices.

The project also demonstrated that establishing a **circular economy system** for electronics is both feasible and beneficial. Instead of viewing e-waste as garbage, it should be seen as a **resource loop** — where components are reused, metals are recovered, and harmful materials are neutralized.

Implementing a structured e-waste policy on campus can reduce the institution’s carbon footprint, minimize pollution, and save money through equipment reuse. Moreover, the project created awareness



among students about their personal responsibility toward sustainable electronic consumption.

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## **Recommendations**

1. **Campus E-Waste Collection Center:**  
Set up a permanent unit for storing and sorting old electronic items.
  2. **Authorized Recycler Tie-Up:**  
Partner with government-approved recyclers for periodic pickup and recycling.
  3. **Reuse and Repair Workshops:**  
Encourage students to refurbish old systems for use in labs or community outreach.
  4. **E-Waste Policy:**  
Develop an official campus-level e-waste management policy in line with CPCB and MoEFCC rules.
  5. **Awareness Drives:**  
Conduct regular sessions on circular economy and e-waste handling practices.
  6. **Digital Tracking System:**  
Introduce QR-based tagging of disposed electronics to track recycling flow.
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## **Limitations**

- Limited sample size restricted total waste estimation.
- Some departments could not participate due to ongoing academic work.
- Lack of access to professional weighing or testing tools for precise measurement.

- Budget constraints prevented bulk transportation to recycling centers.

Despite these limitations, the model successfully demonstrated a small-scale circular economy framework for campus electronics.

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## Future Scope

- Expansion of the model to cover all departments and hostels.
- Establishment of a “**Tech Refurbish Lab**” for repairing discarded electronics.
- Collaboration with NGOs and recyclers for large-scale e-waste collection drives.
- Integration of e-waste topics into curriculum through sustainability courses.
- Development of a **mobile app** for reporting and tracking discarded devices.

This project lays the foundation for a **Zero-E-Waste Campus**, aligning with global sustainable development goals (SDGs) and promoting a technology-driven circular economy.

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## Conclusion

The **E-Waste Estimation & Circular Economy Model** successfully highlighted the volume, types, and management challenges of electronic waste generated within SR University. Through systematic identification, data collection, and classification, our team demonstrated that a significant portion of e-waste can be reused or recycled effectively.

By applying the principles of the circular economy — **reduce, reuse, repair, and recycle** — the project not only mitigates environmental

pollution but also creates pathways for resource conservation and green innovation.

This initiative proved that even small-scale institutional actions can drive meaningful climate impact. Responsible e-waste management is not just about disposal — it's about **rethinking the lifecycle of technology** to build a sustainable digital future.

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## References

1. Central Pollution Control Board (CPCB), *E-Waste Management Rules, 2016 & 2022*.
2. Ministry of Environment, Forest and Climate Change (MoEFCC), *Circular Economy Guidelines for E-Waste*.
3. World Health Organization (WHO), *Health Impacts of Electronic Waste Exposure*.
4. United Nations Environment Programme (UNEP), *Global E-Waste Monitor, 2024*.
5. Intergovernmental Panel on Climate Change (IPCC), *Climate Implications of Resource Recovery, 2021*.
6. National Green Tribunal (NGT), *Reports on E-Waste Regulation, 2023*.
7. The Hindu & Down To Earth, *Articles on E-Waste Recycling Practices in India*.
8. SR University Campus Data Survey, *E-Waste Estimation Study, 2025*.

